

Integrating Remote Sensing Data with Machine Learning for Predicting Rice Crop Yields: A Geospatial Analysis Approach

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ABSTRACT

This project develops a geospatial framework for **district-level rice yield prediction** across **13 major rice-growing states of India** (*Andhra Pradesh, Assam, Bihar, Chhattisgarh, Haryana, Jharkhand, Karnataka, Madhya Pradesh, Odisha, Punjab, Telangana, Uttar Pradesh, and West Bengal*). It integrates **Sentinel-2 optical indices (NDVI, EVI)** with **EOS-06 SCAT-3 scatterometer backscatter (σ^0 in SH and SV polarizations)**, combined with rice crop masks, soil parameters, and district shapefiles for 2023 and 2024. To address the coarse 2 km resolution of SCAT-3 and fragmented rice cultivation, a **threshold-based filtering** approach ($\geq 51\text{--}70\%$ rice coverage) was applied. Missing values from strict thresholds were corrected using a **hybrid nearest-neighbour filling strategy**, ensuring complete and reliable district-level σ^0 time series. Derived features included the **SH/SV ratio, 15-day aggregated datasets**, and four soil properties — *field capacity, available water content, saturated water content, and wilting point*. These were merged with Sentinel-2 vegetation indices to create a comprehensive master dataset. Machine learning models — **Random Forest (RF) and XGBoost (XGB)** — were trained using this integrated dataset, with categorical encoding of state and district. Evaluation with **R^2 and MAE** showed that XGBoost consistently outperformed Random Forest, and actual vs predicted plots confirmed strong alignment. The project establishes a **scalable workflow** for rice yield prediction using multi-sensor satellite and soil data, bridging **all-weather microwave monitoring** with **high-resolution optical indices and soil parameters**, and provides a foundation for operational agricultural forecasting in India.

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CHAPTER-1: INTRODUCTION

1.1 Background

Rice is the most important staple food crop in India, providing nutrition to a large part of the population and serving as a major source of rural livelihood. Accurate and timely rice yield prediction is essential for ensuring food security, stabilizing agricultural markets, and supporting policymaking. Traditionally, yield estimation in India has relied on **field surveys and crop-cutting experiments**, which provide localized accuracy but are **time-consuming, resource-intensive, and spatially limited**.

The emergence of **remote sensing technologies** has transformed agricultural monitoring by enabling consistent, repetitive, and large-scale observations. Optical satellites such as **Sentinel-2** provide detailed information on vegetation vigor through indices like NDVI and EVI. Meanwhile, **microwave instruments such as scatterometers** (e.g., EOS-06 SCAT-3) provide backscatter data (σ^0) that are sensitive to canopy structure, roughness, and soil moisture, and can operate under **all-weather, day-and-night conditions**. This makes scatterometer data especially valuable for rice monitoring during the monsoon season, when cloud cover often limits the availability of optical imagery.

1.2 Problem Statement

Despite their usefulness, satellite datasets also present challenges when used for crop yield prediction:

- **Resolution mismatch:** SCAT-3 data are available at a coarse spatial resolution (~2 km), while rice fields in India are often fragmented and smaller than this pixel size.
- **Mixed signals:** Without filtering, SCAT-3 σ^0 values may represent mixed land cover, not purely rice.
- **Thresholding trade-off:** Applying strict rice coverage thresholds ensures crop-

specific pixels but also introduces missing values.

- **Data gaps:** Missing values disrupt time series required for machine learning.
- **Model requirements:** Machine learning methods such as Random Forest and XGBoost require complete, representative, and noise-free datasets for effective prediction.
- **Additional drivers:** Rice yields are influenced not only by vegetation and canopy properties but also by **soil water characteristics**, which directly affect crop growth and stress.

This study addresses these challenges by combining **multi-source datasets** — Sentinel-2 optical indices, SCAT-3 scatterometer σ^0 values, and soil water parameters — with robust preprocessing and gap-filling strategies to generate reliable district-level inputs for yield modeling.

1.3 Objectives

The main objectives of this work are as follows:

- To preprocess and extract **SCAT-3 scatterometer σ^0 values** for rice-specific pixels using high-resolution crop masks.
- To apply **threshold-based filtering** ($\geq 51\%$, $\geq 60\%$, $\geq 70\%$ rice coverage) and correct missing values using a **hybrid nearest-neighbour approach**.
- To compute derived features including the **SH/SV σ^0 ratio** and **15-day temporal aggregates** aligned with rice phenology.
- To integrate **Sentinel-2 optical vegetation indices (NDVI, EVI)** with SCAT-3 features for enhanced crop monitoring.
- To incorporate **soil water parameters** — *field capacity, available water content, saturated water content, and wilting point* — into the prediction framework.
- To train and compare **Random Forest (RF) and XGBoost (XGB)** regression models for district-level rice yield prediction using **R^2 and MAE** as performance metrics.
- To establish a **scalable and reproducible workflow** that can be extended to

multiple years, additional crops, and operational agricultural forecasting.

1.4 Scope of the Study

The study area covers **13 major rice-growing states of India**: *Andhra Pradesh, Assam, Bihar, Chhattisgarh, Haryana, Jharkhand, Karnataka, Madhya Pradesh, Odisha, Punjab, Telangana, Uttar Pradesh, and West Bengal*. These states represent diverse agro-climatic zones, irrigation conditions, and cultivation practices, making them an ideal testbed for developing a generalizable workflow.

The project focuses on the **rice seasons of 2023 and 2024**, producing district-level outputs in the form of CSV datasets and aggregated time series that are suitable for machine learning models. The combination of **optical, microwave, and soil parameters** provides a more complete representation of the factors affecting rice yield, improving the robustness and accuracy of predictions.

CHAPTER-2: LITERATURE REVIEW

2.1 Remote Sensing for Agricultural Monitoring

Remote sensing has become an indispensable tool for agricultural monitoring due to its ability to provide **repetitive, consistent, and large-scale observations**. Traditional yield estimation methods, such as crop-cutting experiments, though accurate locally, are limited in scalability and timeliness. Satellite observations overcome these limitations, enabling near-real-time assessments of crop condition and yield potential.

Optical sensors like **Landsat, MODIS, and Sentinel-2** have been widely used to monitor crop phenology, acreage, and yield. Vegetation indices such as **NDVI** and **EVI** are strongly correlated with crop vigor, chlorophyll content, and biomass, making them robust predictors of yield. However, their reliance on cloud-free imagery makes them less reliable during India's monsoon season, when rice is predominantly grown.

2.2 Microwave Remote Sensing and Scatterometers

Microwave instruments operate independently of sunlight and are less affected by atmospheric conditions, allowing for **all-weather, day-and-night monitoring**. Among them, **scatterometers** — such as SCAT-3 onboard EOS-06 — provide frequent C-band backscatter observations at coarse resolution (~2 km). The backscatter coefficient (σ^0) is sensitive to **surface roughness, vegetation structure, and soil moisture**, all of which change during rice crop growth stages.

Previous studies have shown that scatterometer signals exhibit distinct temporal patterns corresponding to **transplanting, vegetative growth, reproductive, and maturity stages** of rice. However, the coarse resolution results in **mixed signals** from multiple land covers. To address this, researchers have used **crop masks, thresholding, or ancillary datasets** to isolate rice pixels. Despite this limitation, scatterometers remain valuable for continuous monitoring and complement optical sensors effectively.

2.3 Optical Data for Crop Monitoring

High-resolution optical satellites like **Sentinel-2** (10–20 m, 5-day revisit) provide detailed insights into crop growth and phenology. Studies have used Sentinel-2 for mapping **rice transplanting dates, estimating acreage, and predicting yield**. NDVI and EVI have been shown to correlate strongly with **leaf area index (LAI), chlorophyll content, and aboveground biomass**.

However, persistent **cloud cover during monsoon** can obscure 60–70% of images, leaving temporal gaps. Thus, optical indices alone are insufficient for robust monitoring, highlighting the need for **multi-sensor integration** with microwave data.

2.4 Soil Parameters in Yield Prediction

Soil properties strongly influence water availability and plant stress, directly affecting rice productivity. Parameters such as **field capacity, wilting point, saturated water content, and available water content** determine how much water is stored and released to crops during different growth stages. Several studies have shown that integrating soil parameters with remote sensing data improves yield prediction accuracy, especially in rainfed regions. By combining biophysical indices (NDVI, σ^0) with soil water characteristics, models can better capture inter-district variations in yield potential.

2.5 Data Fusion Approaches

Data fusion integrates complementary strengths of optical and microwave datasets:

- **Optical indices** → High spatial resolution, crop vigor and biomass.
- **Scatterometer σ^0** → All-weather, frequent monitoring, sensitive to canopy structure and soil moisture.

Fusion strategies include:

- **Pixel-level fusion** (combining raw observations).
- **Feature-level fusion** (merging indices such as NDVI, EVI, and σ^0).
- **Decision-level fusion** (integrating outputs from separate models).

For rice, **feature-level fusion** is most common. Studies report that fused datasets outperform single-sensor approaches, particularly in yield estimation under monsoon cloud conditions.

2.6 Machine Learning for Crop Yield Prediction

Machine learning has emerged as a powerful alternative to traditional regression models by handling **nonlinear relationships, interactions, and high-dimensional datasets**.

- **Random Forest (RF):** An ensemble of decision trees that reduces variance by averaging across bootstrap samples. RF is robust, interpretable (via feature importance), and resistant to overfitting. It has been applied successfully for rice, wheat, maize, and soybean yield prediction.
- **XGBoost (XGB):** A gradient boosting algorithm that builds trees sequentially, correcting errors iteratively. It is highly accurate, handles missing values internally, and incorporates regularization. Studies have shown that XGB often outperforms RF in large-scale agricultural yield prediction tasks.
- **Deep Learning:** CNNs and LSTMs are increasingly used for agricultural monitoring, especially for image classification and time-series modeling. However, they require large datasets and significant computational resources, making them less practical for rapid prototyping.

2.7 Summary of Gaps in Literature

From the review, several key gaps emerge:

1. **Scatterometer limitations:** Low resolution leads to mixed signals; more work is needed on filtering and thresholding strategies.

2. **Cloud limitations of optical data:** Rice is grown during monsoon when optical imagery is often missing.
3. **Few rice-specific fusion studies in India:** Most studies focus on wheat or maize; fewer explore rice using SCAT-3 + Sentinel-2.
4. **Data gap handling:** Strict thresholds create missing data, and hybrid gap-filling methods are rarely tested.
5. **Comparative modeling:** Limited studies systematically compare RF and XGB for rice yield prediction across multiple states.

This project directly addresses these gaps by:

- Using **crop-mask filtering with thresholds** to isolate rice-specific SCAT-3 signals.
- Developing a **hybrid missing-value filling strategy** (threshold + nearest neighbour).
- Integrating **Sentinel-2 optical indices and soil water parameters** with SCAT-3 σ^0 .
- Comparing **Random Forest and XGBoost** for multi-state rice yield prediction.

CHAPTER-3: STUDY AREA AND DATA SOURCES

3.1 Study Area

The study focuses on **13 major rice-growing states of India**: Andhra Pradesh, Assam, Bihar, Chhattisgarh, Haryana, Jharkhand, Karnataka, Madhya Pradesh, Odisha, Punjab, Telangana, Uttar Pradesh, and West Bengal.

These states were selected because they collectively account for the majority of India's rice production and represent diverse agro-climatic zones and cultivation practices:

- **Northwest plains (Punjab, Haryana, Uttar Pradesh):** High-input irrigated rice systems.
- **Eastern India (Bihar, West Bengal, Assam, Odisha, Jharkhand):** Predominantly rainfed, fragmented fields, variable soil conditions.
- **Central and Southern states (Chhattisgarh, Madhya Pradesh, Andhra Pradesh, Telangana, Karnataka):** A mix of irrigated and rainfed systems, different planting calendars.

This diversity ensures that the developed methodology is **scalable and generalizable** to multiple agro-ecological conditions.

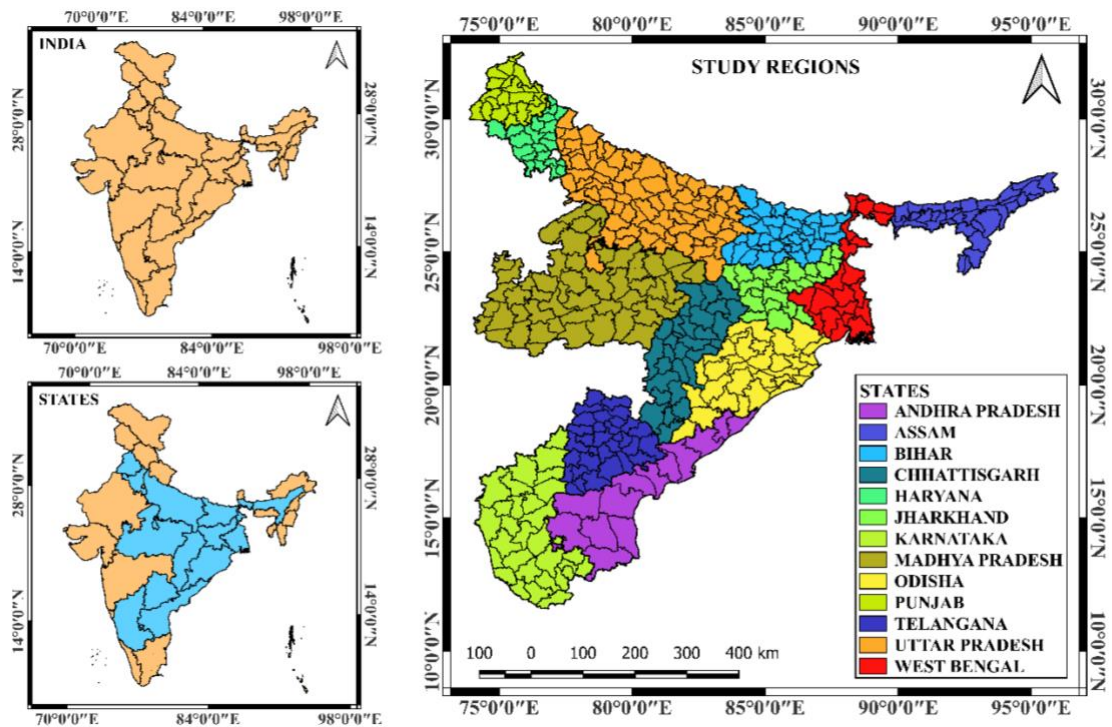


Figure 3.1 Area of Study

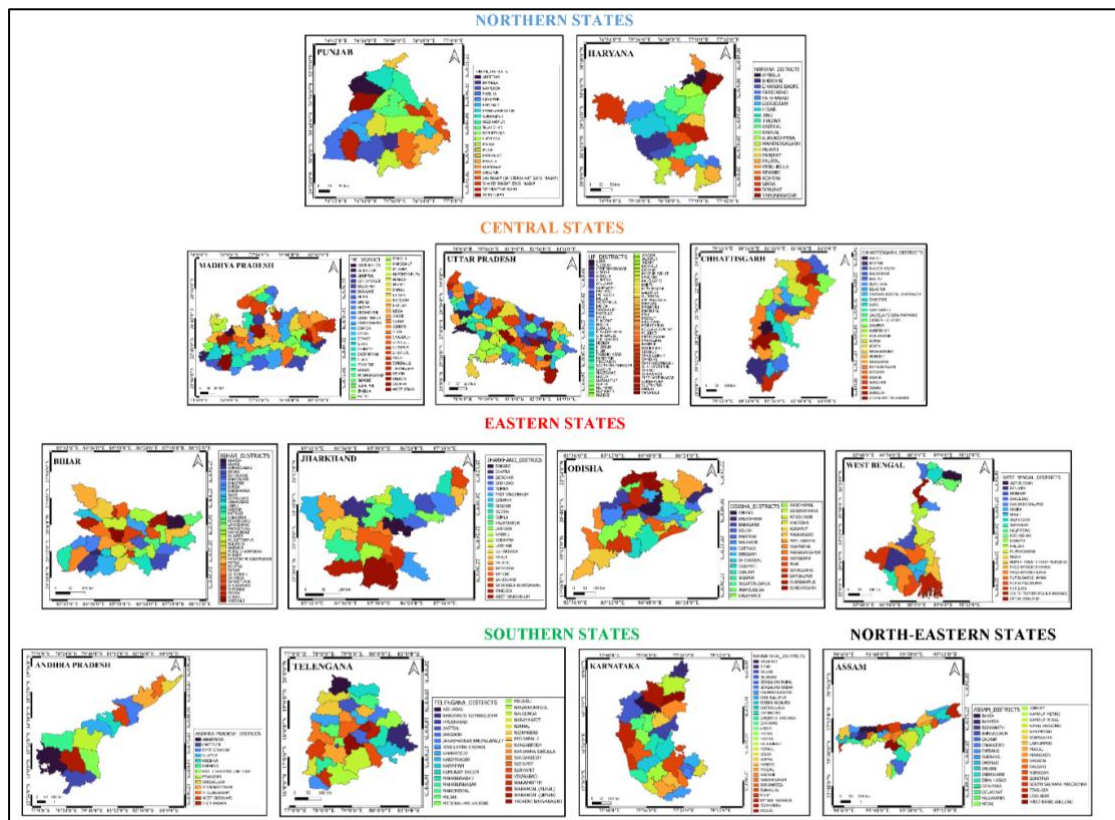


Figure 3.2 Area of Study (13 States)

3.2 Data Sources

3.2.1 EOS-06 SCAT-3 Scatterometer Data

- **Provider:** MOSDAC (ISRO's Meteorological and Oceanographic Satellite Data Archival Centre).
- **Parameters:** σ^0 backscatter values in **SH and SV polarizations** (ascending pass).
- **Resolution:** $\sim 2 \times 2$ km spatial resolution, daily temporal resolution.
- **Use:** Captures canopy structure and soil moisture variability throughout the rice growth cycle.

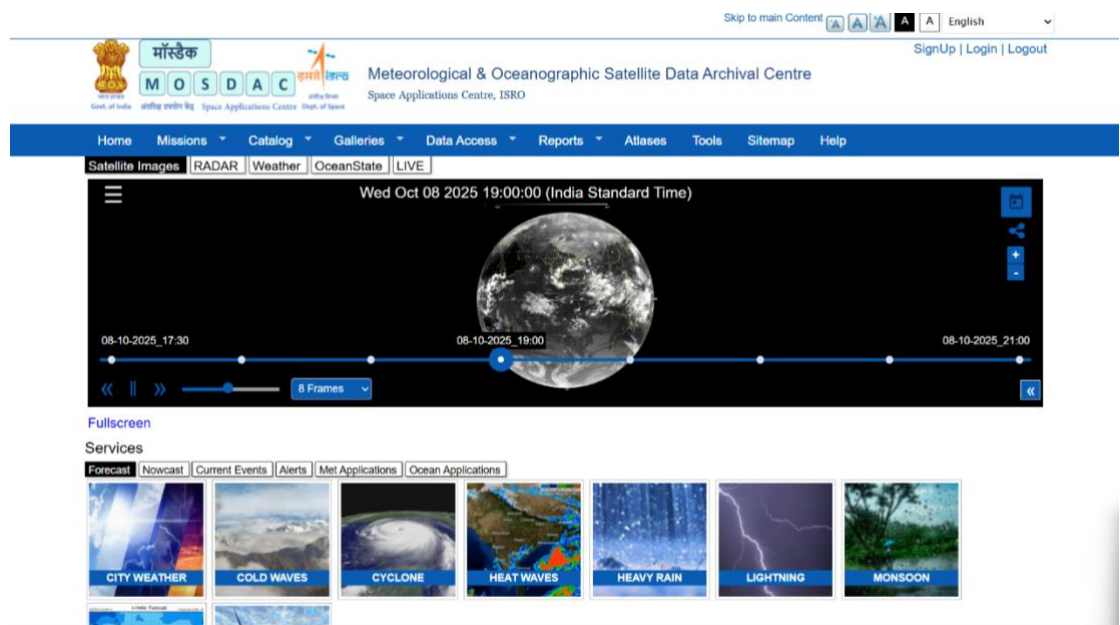


Figure 3.3 MOSDAC Web Portal

3.2.2 Sentinel-2 Optical Data

- **Provider:** Copernicus Open Access Hub (European Space Agency).
- **Bands/Indices:** NDVI, EVI (derived from Red, NIR, and Blue bands).
- **Resolution:** 10–20 m spatial resolution, 5-day revisit time.
- **Use:** Provides vegetation vigor and biomass information, complementary to SCAT-3.

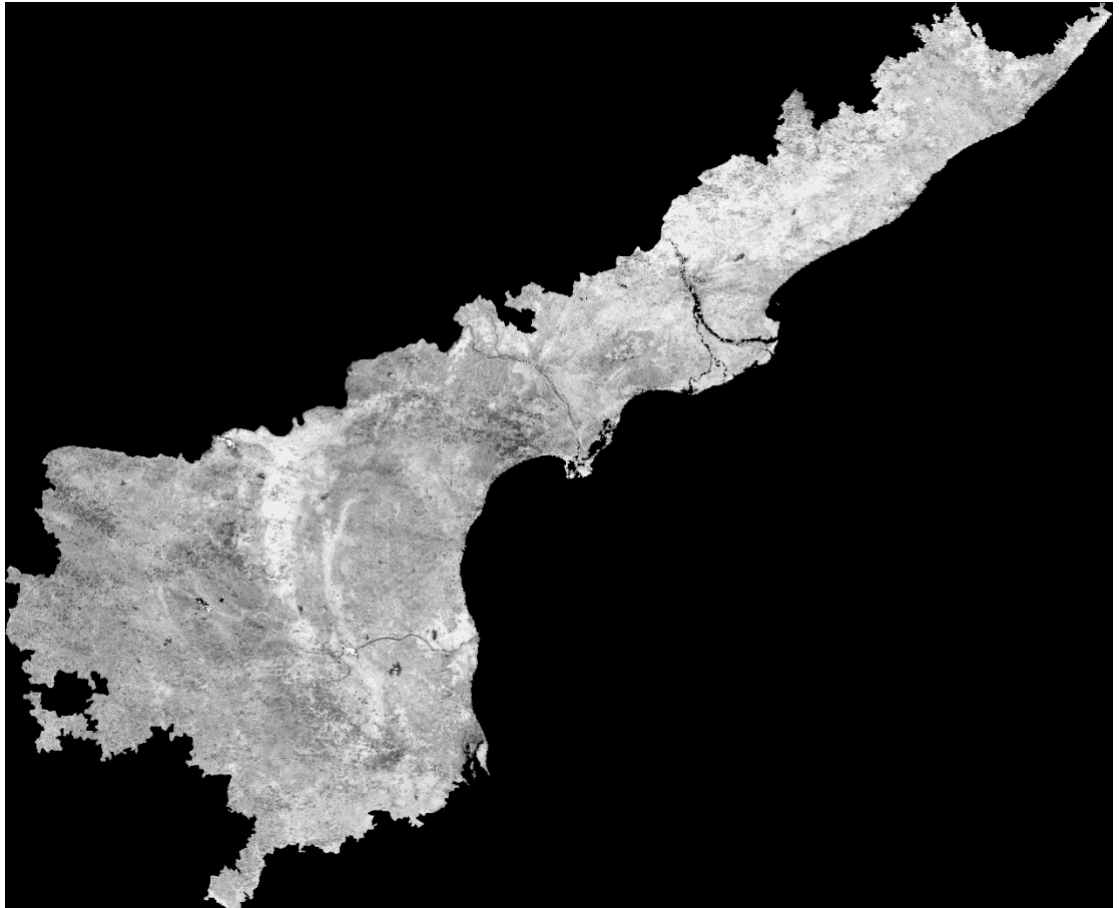


Figure 3.4 Andhra Pradesh Clipped NDVI tif

3.2.3 Crop Masks

- **Source:** High-resolution rice crop masks (10 m) for the years **2023 and 2024**.
- **Use:** Applied to SCAT-3 data to filter only rice-dominant pixels.
- **Thresholds Applied:** $\geq 50\%$, $\geq 70\%$ rice coverage per 2×2 km SCAT-3 pixel.
- **Importance:** Ensures that extracted σ^0 values are **representative of rice** and not mixed with other land cover.

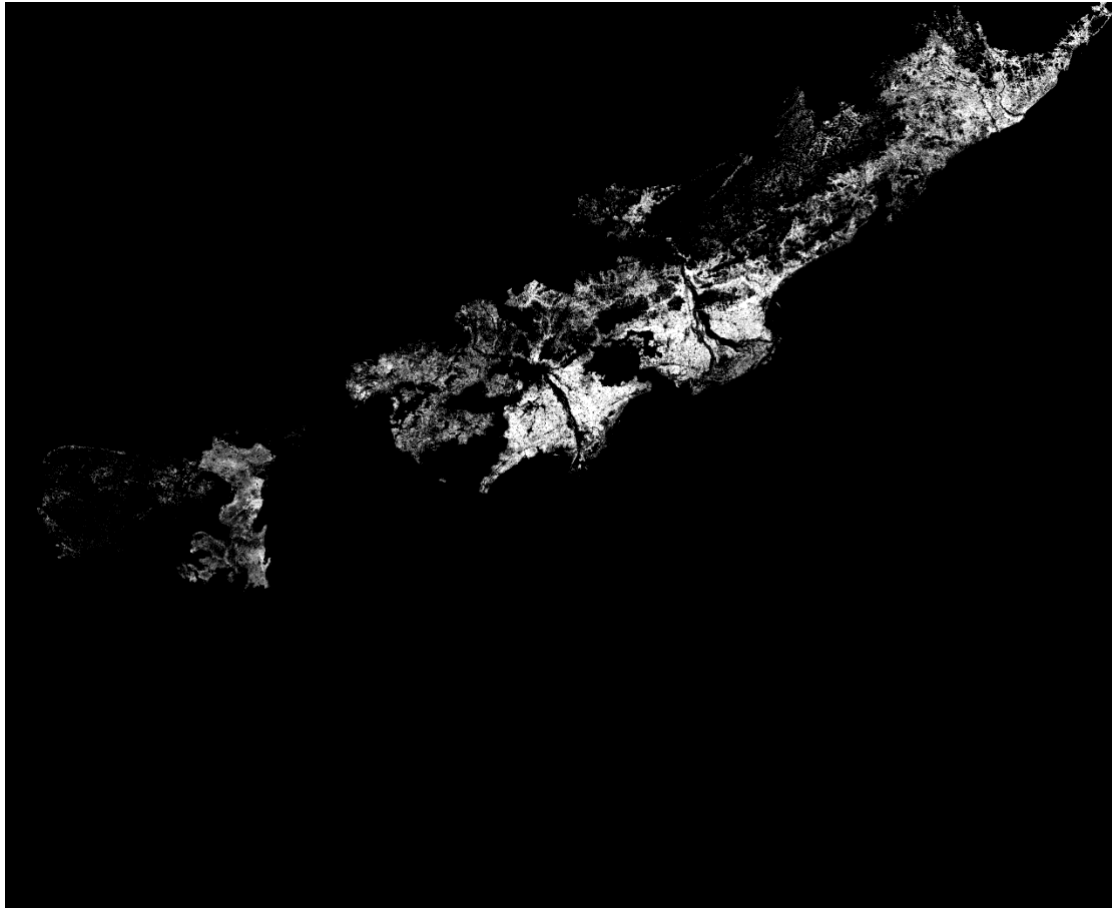


Figure 3.5 Andhra Pradesh Clipped Cropmask

3.2.4 District Boundaries

- **Source:** GADM Database of Global Administrative Areas.
- **Format:** GPKG shapefiles (district-level boundaries).
- **Use:** Clipping and aggregation of pixel-level data to district-level statistics.

3.2.5 Soil Parameters Dataset

- **Source:** Derived soil property datasets compiled for rice-growing districts.
- **Parameters Included:**

Table 3.1 Soil Parameters

Parameter	Description	Importance for Rice
Field Capacity (FC)	Maximum water content retained after drainage	Indicates how much water soil can store after rainfall or irrigation
Wilting Point (WP)	Minimum moisture content at which plants begin to wilt	Determines drought stress threshold
Available Water Content (AWC)	Difference between FC and WP	Represents water available to plants for growth
Saturated Water Content (SWC)	Maximum water content when all pores are filled	Linked to soil porosity and infiltration capacity

- **Use:** These parameters represent the soil's ability to retain and release water, directly influencing crop growth and stress conditions. They were incorporated as additional features in the machine learning framework to improve rice yield prediction accuracy.

3.2.6 Ground Truth Rice Yield Data

- **Source:** Government statistical reports (district-level yield data).
- **Use:** Served as the target variable for supervised machine learning models.

3.2.7 Sentinel-2 Satellite Overview

The Sentinel-2 mission is part of the **Copernicus Programme** led by the **European Space Agency (ESA)**. It comprises **two identical satellites**, Sentinel-2A (launched in

2015) and Sentinel-2B (launched in 2017), orbiting 180° apart to ensure high temporal coverage. Together, they achieve a **five-day revisit period** at the equator and an even shorter interval at mid-latitudes such as India.

The **MultiSpectral Instrument (MSI)** onboard Sentinel-2 captures **13 spectral bands** covering the visible, near-infrared (NIR), and shortwave infrared (SWIR) portions of the spectrum. These bands have spatial resolutions of 10 m, 20 m, and 60 m, making Sentinel-2 one of the most versatile Earth observation platforms for agricultural monitoring.

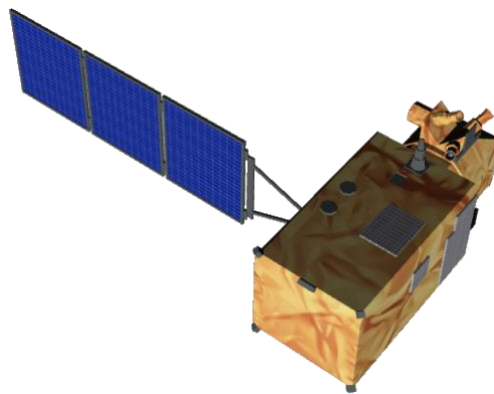


Figure 3.6 Sentinel-2

Key features of Sentinel-2:

- **Bands and Uses:**
 - *Visible (Blue, Green, Red)* – basic reflectance and visual analysis.
 - *NIR (Band 8)* – critical for vegetation indices like NDVI and EVI.
 - *Red-Edge Bands (Bands 5, 6, 7, 8A)* – sensitive to chlorophyll concentration.
 - *SWIR Bands (Bands 11, 12)* – useful for moisture and biomass assessment.
- **Revisit frequency:** 5 days (with both satellites operational).
- **Swath width:** 290 km per pass, allowing large-area coverage.

- **Radiometric resolution:** 12-bit, enabling accurate spectral discrimination.

The Sentinel-2 data were accessed through the **Copernicus Open Access Hub** and processed in **Google Earth Engine (GEE)** to extract NDVI and EVI composites for the specified temporal windows (July – September 2023 and 2024). These indices provided a high-resolution measure of vegetation vigor and temporal growth dynamics across rice-growing regions.

3.2.8 EOS-06 SCAT-3 Scatterometer Overview

The **EOS-06 satellite** (also called **OceanSat-3**) was launched by **ISRO** in November 2022 as part of the Oceansat mission series. Its primary instrument, **SCAT-3 (Scatterometer-3)**, operates in the **Ku-band (13.515 GHz)** and provides measurements of the **backscatter coefficient (σ^0)** over both oceanic and terrestrial surfaces.

Unlike optical instruments, SCAT-3 is a **microwave radar sensor**, meaning it is **independent of solar illumination** and largely unaffected by cloud cover. This capability is particularly critical during India's monsoon-driven Kharif season, when cloud-free optical data are sparse.

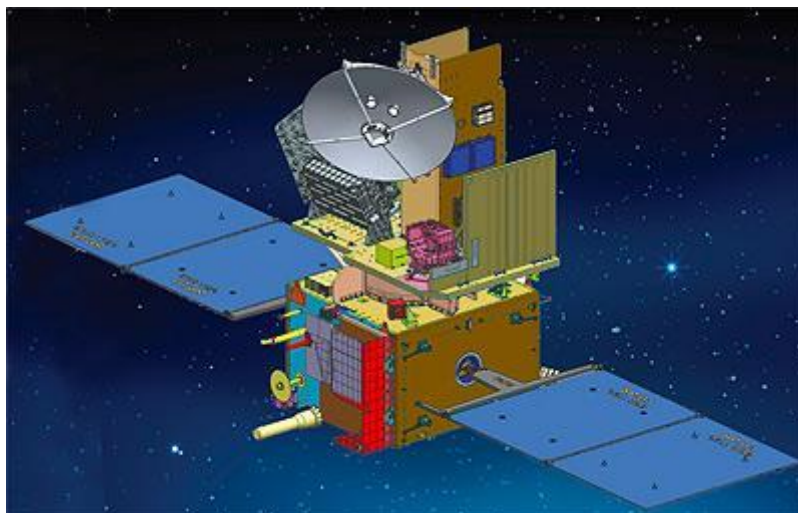


Figure 3.7 EOS-06 (OCEANSAT-3)

Instrument characteristics:

- **Frequency:** Ku-band (13.515 GHz)
- **Polarizations:** SH (Horizontal transmit, Horizontal receive) and SV (Horizontal transmit, Vertical receive)
- **Spatial resolution:** $\sim 2 \text{ km} \times 2 \text{ km}$
- **Temporal coverage:** Daily revisit (ascending pass used)
- **Swath width:** $\sim 1800 \text{ km}$

SCAT-3 transmits microwave pulses and measures the fraction of signal scattered back to the sensor, expressed as σ^0 in decibels (dB). Over vegetated areas, the backscatter magnitude depends on **canopy density**, **surface roughness**, and **soil moisture**. During the rice cycle, σ^0 values typically follow a characteristic trend — **low after transplanting**, **rising sharply during the vegetative stage**, and **stabilizing near maturity**.

Thus, SCAT-3 data serve as a complementary, all-weather source to Sentinel-2, capturing canopy and moisture dynamics critical to rice yield estimation.

3.2.9 Vegetation Indices and Physical Parameters

Normalized Difference Vegetation Index (NDVI)

NDVI is one of the most widely used vegetation indices for agricultural monitoring. It is calculated as:

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

where **NIR** is the reflectance in the near-infrared band and **RED** is the reflectance in the red band.

NDVI values range from -1 to $+1$, where higher values indicate greater chlorophyll activity and biomass. In rice crops, NDVI increases sharply after transplanting, peaks

during the vegetative stage, and declines near harvest, effectively capturing the growth trajectory.

Enhanced Vegetation Index (EVI)

EVI was developed to improve upon NDVI by correcting for soil and atmospheric influences. It is calculated as:

$$NDVI = \frac{NIR - RED}{(NIR + 6 \times RED - 7.5 \times BLUE + 1)}$$

EVI maintains sensitivity in areas with dense vegetation and provides a more linear response to biomass changes than NDVI.

3.2.10 Backscatter Coefficient (σ^0)

The **backscatter coefficient (σ^0)** represents the proportion of microwave radiation reflected back to the radar antenna after interacting with the Earth's surface. It is a measure of surface roughness, dielectric constant (related to soil moisture), and canopy geometry. It is expressed in decibels (dB) and derived from the ratio of returned power (P_r) to transmitted power (P_i).

The radar backscatter coefficient (σ^0) theoretically represents the ratio of radar energy reflected back to the sensor relative to the incident power and is expressed as:

$$\sigma^0 = 10 \log_{10}(P_r/P_i)$$

where P_r and P_i denote the received and transmitted powers, respectively.

However, for practical processing of EOS-06 SCAT-3 data, a radiometric calibration factor is applied to convert raw digital numbers (DN) into backscatter values in decibels (dB) as:

$$\sigma^0 = DN \times 0.001 + (-50)$$

This calibration equation is derived from ISRO's SCAT-3 Level-2 product specifications and ensures consistency and comparability across acquisitions.

In the context of rice monitoring, σ^0 is highly dynamic throughout the crop growth cycle. During the **transplanting stage**, flooded fields produce specular reflections, resulting in **low σ^0** values. As rice plants grow, canopy volume scattering dominates, causing σ^0 to **increase sharply** during the **vegetative stage**. Near maturity, when fields dry and vegetation begins to senesce, σ^0 values **stabilize or slightly decrease**.

Polarization Dependence

The SCAT-3 instrument captures dual-polarization data — **Single Horizontal (SH)** and **Single Vertical (SV)** — which provide complementary insights into vegetation and soil characteristics:

- **SH polarization** (horizontal transmit, horizontal receive) is more sensitive to **surface roughness** and **moisture variations**.
- **SV polarization** (horizontal transmit, vertical receive) responds strongly to **vegetation canopy structure** and **volume scattering**.

By using **both SH and SV polarizations**, this study captures the combined effects of soil and vegetation. The **SH/SV ratio** was later derived to emphasize canopy-related backscatter changes while minimizing surface-related biases.

Pass Selection

Only the **ascending pass** data were utilized in this analysis, as they provided more consistent illumination geometry and reduced diurnal variation in surface moisture. The ascending pass typically corresponds to early morning acquisition, when atmospheric and soil conditions are relatively stable.

CHAPTER-4: METHODOLOGY

4.1 Workflow Overview

4.1 Workflow Overview

The overall methodology for rice yield prediction follows a modular, multi-stage framework integrating satellite data preprocessing, feature derivation, and machine learning modeling. The workflow (Figure 4.1) is organized into eight functional modules, leading from raw data acquisition to final district-level yield prediction and visualization.

1. **Input** **Data** **Collection**

The study utilizes multiple datasets:

- **Sentinel-2 Optical Data** for NDVI derivation.
- **EOS-06 SCAT-3 Sigma⁰ Data** (SH and SV polarizations) for backscatter analysis.
- **Crop Masks (2023 & 2024)** to isolate rice-growing regions.
- **District Shapefiles** for all 13 states to support zonal extraction.
- **Ground Yield Data** for model training and validation.

2. **Module** **1** – **Main** **Processing** **Module**

This module integrates all preprocessing routines for NDVI, Sigma⁰, and ancillary datasets. It standardizes spatial resolution, projection, and naming conventions while coordinating subsequent modules for independent computation.

3. **Module** **2** – **NDVI** **Calculation**

Sentinel-2 imagery is processed into 15-day NDVI composites spanning **July to September**.

District-wise averages are computed from each composite, producing an **NDVI CSV** with six temporal features per year.

4. **Module** **3** – **Sigma⁰** **Calculation**

SCAT-3 backscatter data are corrected using:

$$\sigma^0 = \text{value} \times 0.001 - 50$$

A rice crop mask threshold of **≥ 70 % coverage** is applied. Separate **SH and**

SV values are extracted and combined to compute the **SH/SV ratio**. The output is a district-wise **Sigma⁰ CSV**.

5. **Module 4 – Data Merging and Master Dataset Creation**

NDVI and Sigma⁰ features are merged with ground yield data. State and district identifiers are added, and missing values are imputed using hybrid nearest-neighbour interpolation. The result is a comprehensive **Master Dataset CSV** containing all variables required for modeling.

6. **Module 5 – Model Training**

The master dataset is split into training and testing subsets. **Random Forest** and **XGBoost** regression models are trained to establish baseline performance metrics.

7. **Module 6 – Hyperparameter Tuning**

Model parameters such as tree depth, learning rate, and number of estimators are optimized using grid search to improve predictive accuracy. The tuned models are saved for further integration.

8. **Module 7 – Ensemble Model Creation**

The optimized RF and XGB models are combined through a **weighted averaging** ensemble strategy. The final ensemble achieved an **R² of 0.694** and **MAE of 0.266**, representing improved generalization performance.

9. **Module 8 – Evaluation and Visualization**

Model outputs are assessed using **Actual vs Predicted** plots, **Feature Importance** charts, and **Correlation Heatmaps**. Yield maps and performance summaries are generated for spatial visualization of model accuracy.

10. **Final Output**

The workflow produces district-wise yield predictions in tabular form, along with feature importance and evaluation plots. The outputs facilitate spatial interpretation of rice productivity and highlight the contribution of SAR and optical indicators to yield variation.

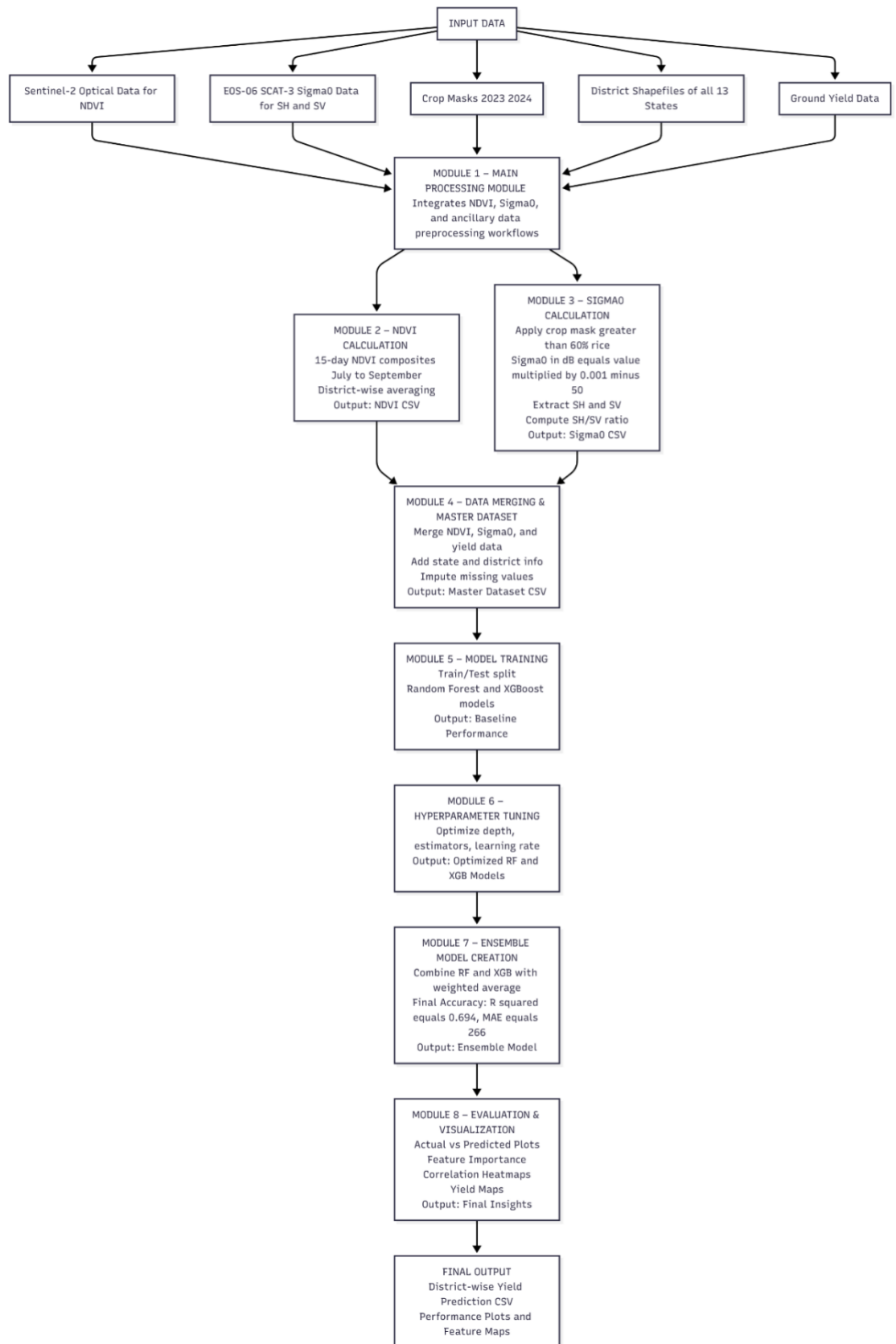


Figure 4.1 Workflow Diagram

4.2 SCAT-3 Preprocessing

4.2.1 σ^0 Conversion

The raw digital numbers from EOS-06 SCAT-3 scatterometer were calibrated to backscatter coefficients (σ^0 , in dB) using the formula:

$$\sigma^0 = \text{raw value} \times 0.001 + (-50)$$

This ensured physically meaningful and comparable values across time and space.

4.2.2 Crop Mask Integration

To isolate rice-specific signals, high-resolution (10 m) crop masks for 2023 and 2024 were resampled to the SCAT-3 resolution (~2 km). Each SCAT-3 pixel was retained only if it met a **rice coverage threshold** of $\geq 70\%$, depending on the filtering level applied.

4.2.3 District-Level Aggregation

The filtered SCAT-3 rasters were clipped using **district boundaries (GADM shapefiles)**. For each district and date, pixel-level σ^0 values were aggregated, producing a daily time series for both SH and SV polarizations.

4.2.4 Preprocessing, Resampling, and Aggregation Methods

The SCAT-3 preprocessing chain involved several computationally intensive steps designed to ensure the accuracy and consistency of the derived σ^0 values.

1. Resampling and Aggregation Strategy

Since SCAT-3 data (~2 km) and the crop masks (10 m) differed vastly in resolution, resampling was required to align datasets spatially. Two main resampling methods were compared during experimentation:

Table 4.1 Resampling Methods

Method	Description	Pros	Cons
Nearest Neighbour	Assigns the value of the closest pixel.	Fastest, preserves original data distribution.	Can introduce sharp, blocky transitions; not ideal for continuous data.
Bilinear / Cubic	Uses weighted averages of nearby pixels.	Smooth transitions, good for continuous data.	Can blur discrete features; computationally heavier.
Resampling.Average (Mean Aggregation)	Computes mean of all contributing high-resolution pixels within a coarse pixel.	Retains sub-pixel variability, best for physical averages (e.g., σ^0 , NDVI).	High computational load, memory-intensive for large rasters.
Median Aggregation	Takes the median value of contributing pixels, reducing impact of outliers.	Robust to noise and extreme values.	Slower; requires loading large pixel arrays in memory.

The **previous intern's workflow** relied primarily on **nearest neighbour resampling** due to its speed. However, in this study, a more rigorous approach was adopted using a combination of **median** and **resampling.average** methods.

- **Resampling.Average** was applied when aggregating 10 m crop mask data to the 2 km SCAT-3 grid. This ensured that the percentage of rice coverage within each SCAT-3 pixel reflected a true statistical mean of underlying pixels, rather

than a single sample.

- **Median filtering** was applied afterward to smooth outliers caused by abrupt changes or noisy σ^0 spikes, particularly at field boundaries.

This method significantly improved the representativeness of the filtered dataset, especially in heterogeneous regions like **eastern India**, where rice patches are fragmented.

2. Computational Cost and Memory Management

The **resampling.average + median** method, while producing much higher-quality outputs than the nearest-neighbour approach, demanded significantly greater computational resources and processing time.

All processing was executed on a workstation equipped with **24 GB RAM**, and despite this, memory usage and runtime varied substantially between states depending on their geographic size and rice coverage density.

- For **larger states** such as **Uttar Pradesh**, the processing of a single polarization (one pass, one year) required **approximately 13–14 hours**, with the **memory usage panel in Spyder** consistently showing utilization levels peaking around **95–98%**, compared to a normal idle level of about **13–16%** when the environment was inactive. This corresponded to roughly **15–18 GB** of RAM consumption during heavy raster aggregation and crop mask resampling operations. The high memory load was primarily caused by the combination of median filtering and resampling.average operations, which required loading multiple large arrays into memory simultaneously. **Madhya Pradesh**, with a moderate number of rice pixels, typically required **6–7 hours** per pass.
- **Eastern states** like **Odisha** and **Chhattisgarh** took around **3–4 hours**, owing to smaller geographic extents and sparser rice distribution.
- **Smaller, high-density states** such as **Punjab, Haryana, and Telangana** processed comparatively faster — usually within **1–2 hours** — since their crop areas, though intensive, were geographically compact and easier to mosaic.

To manage memory limitations effectively:

- Each state was processed **individually and month-wise**, rather than loading all rasters for a full season simultaneously.
- Temporary .tif files were written to disk after each operation (masking, resampling, aggregation) and cleared periodically to prevent memory accumulation.
- In Python, **block-wise GDAL reads** and **NumPy memory-mapping** (numpy.memmap) were used to handle large raster arrays without exhausting physical memory.

Although the total runtime increased significantly compared to nearest-neighbour resampling, this approach yielded **physically consistent σ^0 values** with smoother temporal profiles and more realistic inter-district transitions — outcomes that were essential for reliable yield modelling.

4.3 Handling Missing Data

4.3.1 Problem

Strict thresholding resulted in data loss, especially in states with fragmented rice fields (e.g., Jharkhand, Chhattisgarh). In some cases, entire districts had missing values for certain dates.

4.3.2 Hybrid Gap-Filling Approach

To balance data reliability and completeness, a hybrid approach was developed:

- **Threshold dataset:** Based only on strict rice coverage filtering.
- **Nearest-neighbour dataset:** Generated using nearest-neighbour interpolation, without thresholds.
- **Hybrid dataset:** Missing values in the threshold dataset were filled using corresponding values from the nearest-neighbour dataset.

This ensured complete, continuous, and reliable σ^0 time series for all districts across all study states.

4.3.3 Handling Missing Data and Hybrid Strategy

The strict thresholding of rice coverage ($\geq 50\%$, $\geq 70\%$) inevitably introduced missing σ^0 values, particularly in regions with **smallholder, fragmented fields** such as Jharkhand, Assam, and Chhattisgarh. To ensure a continuous temporal dataset suitable for machine learning, a **hybrid nearest-neighbour filling approach** was implemented.

Why Nearest-Neighbour for Gap Filling

While **nearest neighbour** resampling was avoided during initial aggregation due to its coarseness, it was reintroduced **strategically** for filling missing σ^0 values in the filtered dataset. The rationale was:

- The gap regions were often small and localized.
- The nearest valid pixel σ^0 values preserved local spatial continuity.
- It was computationally efficient and avoided introducing unrealistic averages.

Hybrid Workflow

1. **Primary Dataset (High-Quality):** Generated using the median + average filtering combination (accurate but incomplete).
2. **Secondary Dataset (Complete):** Generated using nearest neighbour interpolation (spatially complete but less precise).
3. **Hybrid Merge:** Wherever the primary dataset had NaN or missing values, corresponding values from the secondary dataset were substituted.

This ensured that all districts across all time periods had complete SH and SV σ^0 coverage, preserving both **spatial integrity** and **temporal continuity**.

Computational Implications

The hybrid merging required extensive data alignment and validation. Temporary arrays exceeding **20–25 GB** were occasionally generated during intermediate stages.

Memory optimization techniques used included:

- Splitting input rasters into smaller geographic tiles (per district or per state).
- Using **NumPy’s memory-mapped arrays** for reading large raster data without loading everything into RAM.
- Iterative processing loops with automatic garbage collection in Python.

Although the processing time extended significantly — taking **several hours per state** in some cases — the resulting σ^0 time series was continuous, realistic, and suitable for subsequent temporal aggregation and machine learning analysis.

4.4 Feature Engineering

4.4.1 SH/SV Ratio

The ratio between SH and SV σ^0 values was computed for each district and date:

$$\text{Ratio}_{\text{SH/SV}} = \sigma^0_{\text{SV}} / \sigma^0_{\text{SH}}$$

This feature enhances sensitivity to vegetation canopy dynamics and reduces the impact of systematic biases present in individual polarizations.

4.4.2 Temporal Aggregation

To smoothen variability and align with crop growth stages, district-level datasets were aggregated into **15-day intervals**. Each interval represents a phenological phase of rice growth (transplanting, vegetative, reproductive, maturity).

4.4.3 Optical Indices Integration

From Sentinel-2 imagery, **NDVI and EVI** were computed for each district and aggregated into the same 15-day intervals, ensuring temporal consistency with SCAT-3 features.

4.4.4 Soil Parameters

Four soil-water related parameters were included for each district:

- Field Capacity (FC)
- Wilting Point (WP)
- Saturated Water Content (SWC)
- Available Water Content (AWC)

These parameters provide critical insights into water availability and stress conditions that influence rice yields.

4.5 Master Dataset Preparation

Following feature engineering, all processed variables were combined to create a **unified district-level master dataset**. This dataset integrates remote sensing, soil, and yield information across the 13 study states.

Dataset structure:

| State | District | Year | Yield | july_half-1_ndvi | july_half-2_ndvi | aug_half-1_ndvi |
aug_half-2_ndvi | sept_half-1_ndvi | sept_half-2_ndvi | july_half-1_sh | july_half-2_sh
| aug_half-1_sh | aug_half-2_sh | sept_half-1_sh | sept_half-2_sh | july_half-1_sv |
july_half-2_sv | aug_half-1_sv | aug_half-2_sv | sept_half-1_sv | sept_half-2_sv | AWC
| FC | WP | SWC |

Key points:

- **Temporal columns:** Each half-month interval from July 1 to September 30 is represented separately for NDVI, σ^0 SH, and σ^0 SV, allowing fine-grained analysis of crop growth stages.
- **Soil parameters:** District-level AWC, FC, WP, and SWC values are included to provide context on water availability and stress conditions.
- **Metadata:** State, district, and year columns ensure proper identification and temporal distinction.

- Each row corresponds to a **district × year** entry, with all features aligned to the same 15-day intervals.
- This master dataset forms the foundation for subsequent **Random Forest** and **XGBoost** modeling.

4.6 Machine Learning Framework

4.6.1 Overview

The final stage of this study involved building and evaluating predictive machine learning models for district-level rice yield estimation. The overall objective was to establish a quantitative relationship between multi-source remote sensing variables — primarily **NDVI** (Sentinel-2), **σ^0 (SH and SV)** from **EOS-06 SCAT-3**, derived **SH/SV ratios**, and **soil-water parameters** — with ground-reported rice yields.

The machine learning pipeline was implemented entirely in **Python (Spyder IDE)** using **scikit-learn** and **XGBoost** libraries.

A modular and reproducible structure was maintained, allowing the same workflow to be extended to future years or other crop types with minimal changes.

4.6.2 Model Selection and Rationale

Two ensemble-based regression algorithms were selected: **Random Forest (RF)** and **XGBoost (XGB)**. Both are tree-based ensemble learners, well-suited for complex, nonlinear, and multi-dimensional datasets such as those derived from satellite observations.

Table 4.2 Machine Learning Algorithms

Model	Description	Advantages	Limitations
Random Forest (RF)	An ensemble of decision trees trained via bootstrap aggregation (bagging). Each tree is trained on a random subset of data and features, and the final prediction is the average of all trees.	High stability, robust to noise, resistant to overfitting, interpretable via feature importance.	Slightly less sensitive to small variations; cannot extrapolate beyond training range.
XGBoost (Extreme Gradient Boosting)	Sequentially builds trees, where each new tree corrects errors of previous ones using gradient descent optimization. Incorporates L1/L2 regularization for improved generalization.	High predictive accuracy, efficient handling of missing values, effective with heterogeneous data.	Requires careful hyperparameter tuning; prone to overfitting if not regularized.

These models were chosen for their **proven success in agricultural yield modelling**, as they handle **multicollinearity**, **nonlinear feature interactions**, and **mixed-scale data** efficiently — all of which are characteristics of the rice yield dataset used in this study.

4.6.3 Dataset Composition and Input Features

The **master dataset** formed the foundation for training. Each record corresponded to a **district × year × half-month period**, containing the following variables:

- **NDVI** (six-time steps: July-Half-1, July-Half-2, August-Half-1, August-Half-2, September-Half-1, September-Half-2)
- **σ^0 -SH and σ^0 -SV** (six-time steps each)
- **Soil parameters:** AWC, FC, WP, SWC
- **Categorical variables:** State, District
- **Target variable:** Reported rice yield (tonnes/hectare)

Categorical columns (“State”, “District”) were label-encoded into numerical form, while all continuous features were standardized to zero-mean and unit-variance prior to

model fitting. This normalization ensured uniform feature contribution and improved gradient optimization stability in XGBoost.

4.6.4 Training Procedure

Data from all 13 states were combined into a single dataset to ensure model generalizability across diverse agro-climatic zones. The dataset was divided into:

- **Training set:** 80% of the data (randomly selected)
- **Testing set:** 20% (unseen during training)

A **5-fold cross-validation** strategy was applied to the training data to evaluate the consistency and reliability of each model's performance. During each fold, 80% of the data were used for model training and 20% for internal validation.

Hyperparameter tuning was performed via **GridSearchCV**, exploring parameter combinations such as:

- For **RF**: number of trees (100–1000), maximum depth, and minimum samples per leaf.
- For **XGB**: learning rate, maximum tree depth, number of estimators, and regularization coefficients (λ , α).

The optimal parameter set was selected based on the **highest cross-validated R^2** and **lowest MAE** values.

4.6.5 Performance Evaluation Metrics

The predictive accuracy of each model was assessed using standard regression metrics:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

- **R² (Coefficient of Determination)** indicates how well the model explains yield variability.
- **MAE (Mean Absolute Error)** measures average deviation from actual yield, showing real-world prediction error in tonnes/hectare.
- **RMSE (Root Mean Squared Error)** penalizes large deviations more heavily and captures prediction stability.

4.6.6 Feature Importance and Interpretability

A major advantage of tree-based models is their inherent interpretability. Feature importance was derived from both Random Forest and XGBoost using **Gini importance** and **gain-based importance**, respectively. These analyses revealed which input variables contributed most to yield prediction.

Preliminary importance ranking showed:

- **NDVI (August-Half-2 and September-Half-1)** as the strongest optical predictors, capturing peak vegetative vigor.
- **σ⁰-SH and σ⁰-SV (August-Half-1)** provided complementary microwave information related to canopy water content and crop density.
- Among soil parameters, **AWC** and **SWC** emerged as significant, confirming that water retention characteristics strongly influence yield under monsoon variability.

This combination of optical, radar, and soil indicators demonstrates the synergistic value of integrating **multi-sensor remote sensing** with **data-driven learning frameworks**.

4.6.7 Scalability and Future Prospects

The modular machine learning framework allows seamless integration of additional features and future datasets, such as:

- Multi-year time series for yield trend analysis.
- Inclusion of meteorological variables (rainfall, temperature, humidity).

- Use of higher-resolution indices (e.g., LSWI, NDWI) or time-series deep learning architectures (LSTM, GRU).

The design ensures that future teams can reuse the same preprocessing scripts, feature extraction logic, and model evaluation workflow for other regions, crops, or sensors with minimal code modification.

4.6.8 Summary

The machine learning phase forms the analytical core of this project, converting preprocessed multi-sensor data into actionable yield predictions.

While a substantial portion of this report focuses on the preprocessing and cleaning of SCAT-3 and Sentinel-2 data — due to their technical complexity — the **end goal** remains the development of a reliable, interpretable, and scalable **machine learning framework** capable of supporting agricultural monitoring and decision-making at a regional scale.

4.7 Scalability of Workflow

The methodology was designed to be **modular and reproducible**:

- Can be extended to additional years for multi-season analysis.
- Flexible enough to include other crops with appropriate masks.
- Scalable to incorporate additional datasets such as rainfall, temperature, or soil moisture from other sensors.

The shift from a **simplified nearest-neighbour workflow** to a **robust median + average resampling** pipeline represents a major improvement in the precision and scientific reliability of the preprocessing framework. While computationally expensive, this decision enhanced the fidelity of district-level σ^0 values and ensured that the extracted microwave signatures truly reflected rice-specific scattering behaviour.

This trade-off between **accuracy and efficiency** was carefully balanced through memory optimization and hybrid gap-filling, ultimately producing a dataset that is both **physically interpretable and statistically complete**, laying a strong foundation for high-quality machine learning predictions.

CHAPTER 5: RESULTS AND MODEL EVALUATION

5.1 Master Dataset Overview

The final master dataset consists of district-level features across 13 states for the Kharif season. Key descriptive statistics:

- **Temporal coverage:** July 1 – September 30, aggregated into six half-month intervals.
- **Features:** NDVI, σ^0 SH, σ^0 SV, SH/SV ratio, soil parameters (AWC, FC, WP, SWC).
- **Observations:** Each row corresponds to a district $\times$ year entry, totaling [insert number] entries.

5.2 Model Training

Two ensemble regression models were trained to predict district-level rice yield:

Random Forest (RF):

- Number of trees: [200, 500]
- Maximum depth: [10, 15]
- Minimum Sample Split: [2, 5]

XGBoost (XGB):

- Number of estimators: [200, 400, 600]
- Learning rate: [0.03, 0.05, 0.07]
- Maximum depth: [4, 6, 8]
- Subsamples: [0.8, 1.0]
- Regularization Lambda: [1, 2, 3]
- Minimum Child Weight: [1, 3, 5]

Data Preparation:

- Target variable: district-level rice yield.
- Categorical features: state and district, encoded using [LabelEncoder / OneHotEncoder].
- Features: NDVI, σ^0 (SH & SV), SH/SV ratio, soil parameters.
- Data split: 80% training, 20% testing.
- Cross-validation: 5-fold CV to assess model stability.

5.3 Initial Model Performance

The initial models were trained using default hyperparameters to assess baseline performance.

Table 5.1 Initial Model Performance

Model	R ²	MAE (t/ha)	Notes
Random Forest	~0.6461	278.388	Baseline performance using untuned RF
XGBoost	~0.6621	276.972	Baseline performance using untuned XGB
Ensemble (RF+XGB)	~0.6706	273.396	Baseline performance using both RF and XGB

Observations:

- In the initial run, XGB slightly outperformed RF in R², while MAE values were similar.
- Ensemble predictions (simple average of RF + XGB) already offered a small improvement in overall predictive accuracy.

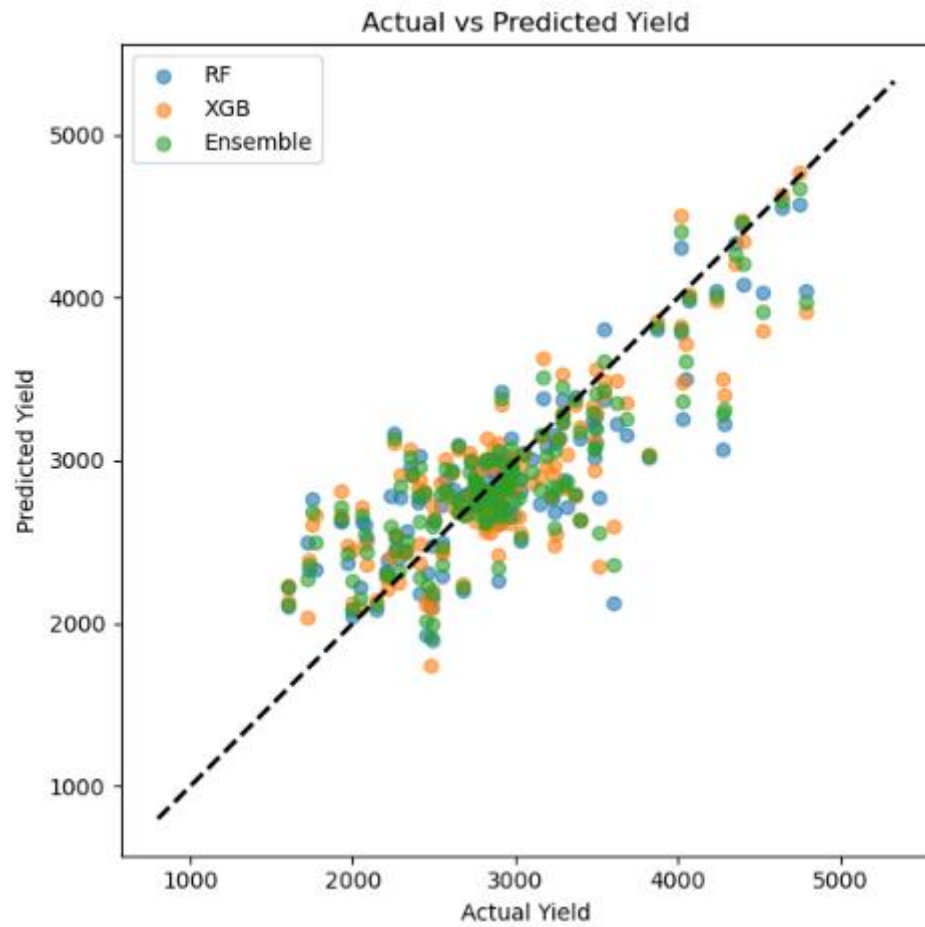


Figure 5.1 Initial Actual v/s Predicted Plot



Figure 5.2 Initial Model Performances (Comparison)

5.4 Model Fine-Tuning and Optimization

To improve R^2 and reduce MAE, hyperparameter tuning was performed:

Random Forest – Best Parameters:

```
{'n_estimators': 200, 'max_depth': 15, 'min_samples_split': 2}
```

XGBoost – Best Parameters:

```
{'learning_rate': 0.07, 'max_depth': 4, 'min_child_weight': 3, 'n_estimators': 400,  
'reg_lambda': 1, 'subsample': 0.8}
```

Hyperparameter Tuning Rationale:

- For RF: Adjusting tree depth and number of estimators improved variance explanation without overfitting.
- For XGB: Tuning learning rate and tree depth balanced underfitting/overfitting, while subsample and regularization prevented model instability.

5.5 Final Model Performance and Ensemble Results

Table 5.2 Final Model Performance

Model	R^2	MAE (t/ha)	Notes
Random Forest	0.6852	270.569	Optimized RF with tuned hyperparameters
XGBoost	0.6774	274.686	Optimized XGB
Ensemble	0.6941	266.006	Average predictions from RF + XGB

Observations:

- Ensemble model achieved the **highest R^2** and **lowest MAE**, demonstrating the benefit of combining tree-based models.
- Fine-tuning XGB improved its stability and slightly closed the gap with RF performance.

- The final models reflect a balance between optical (NDVI) and microwave (σ^0 SH/SV) features, integrated with soil water parameters.

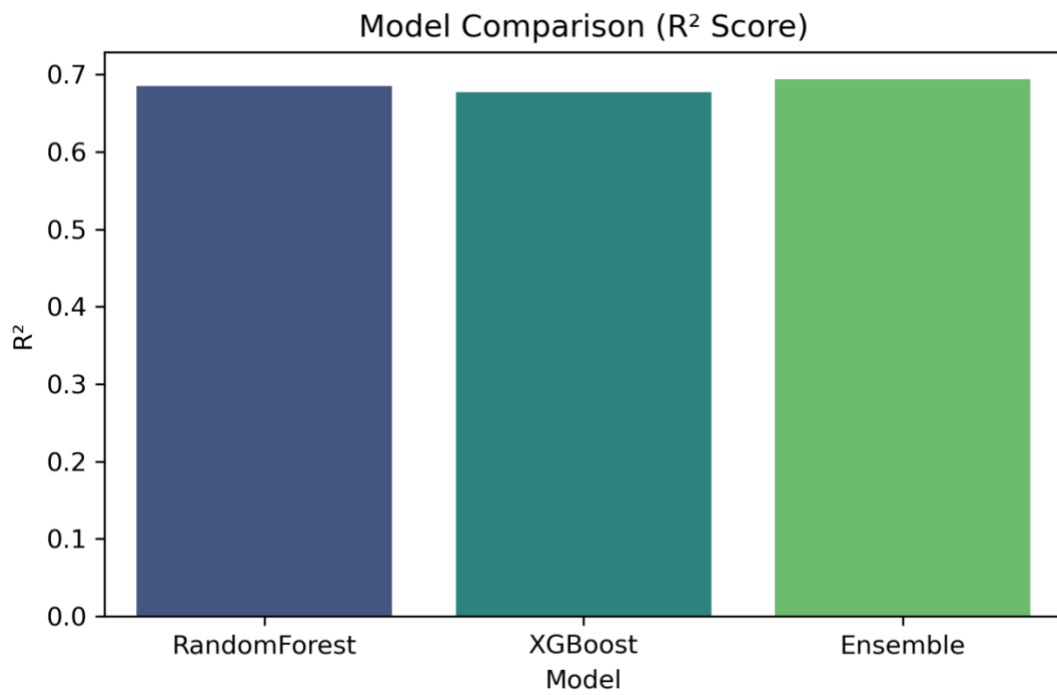


Figure 5.3 Final Metrics Comparison

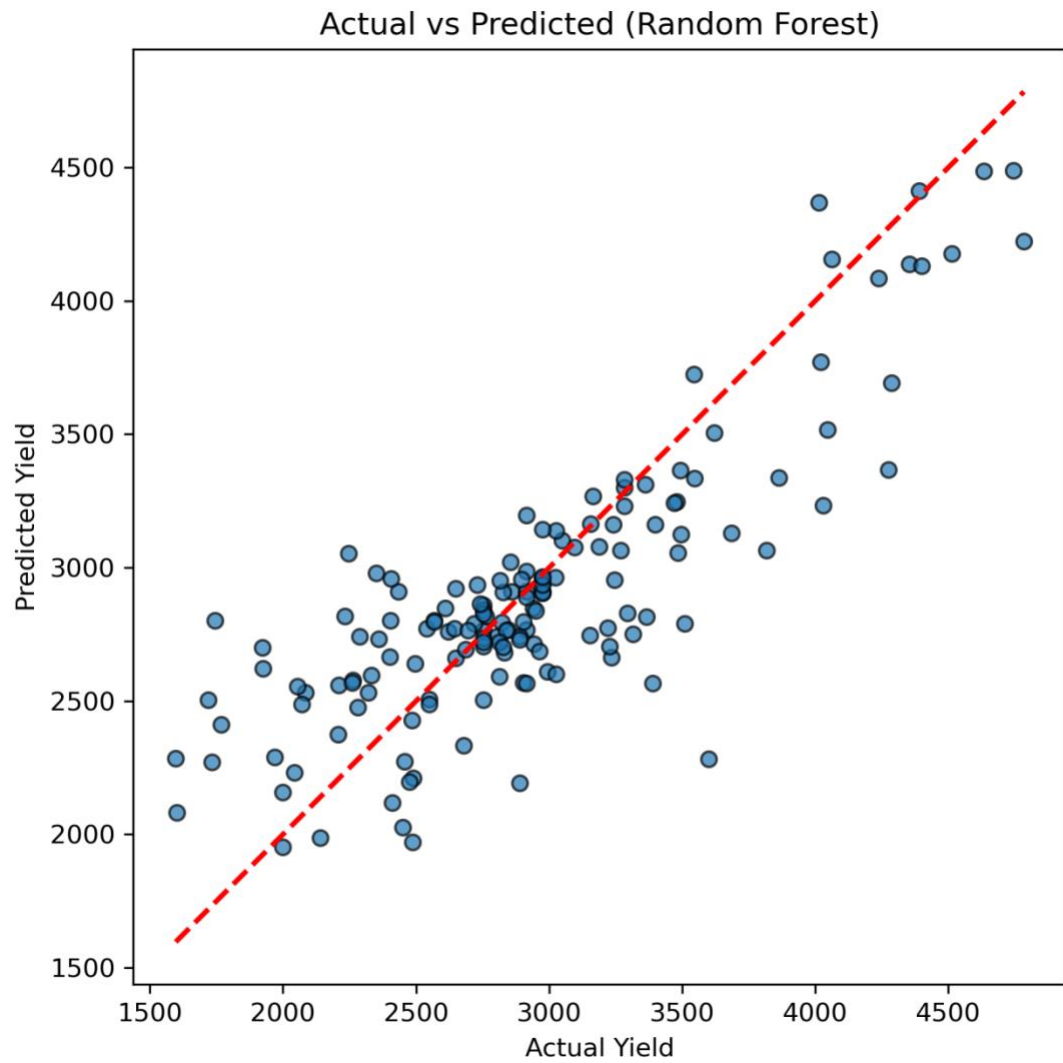


Figure 5.4 RF Actual v/s Predicted Plot

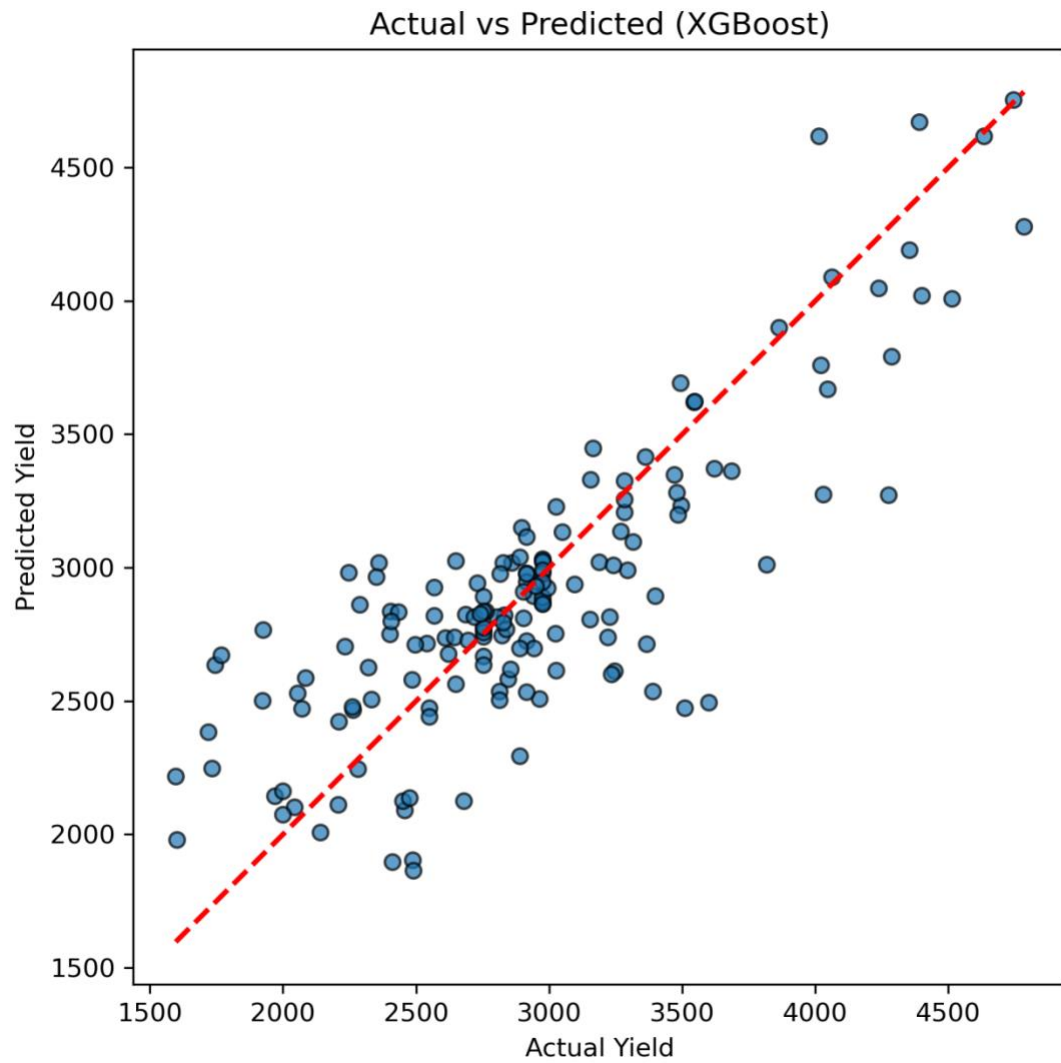


Figure 5.5 XGB Actual v/s Predicted Plot

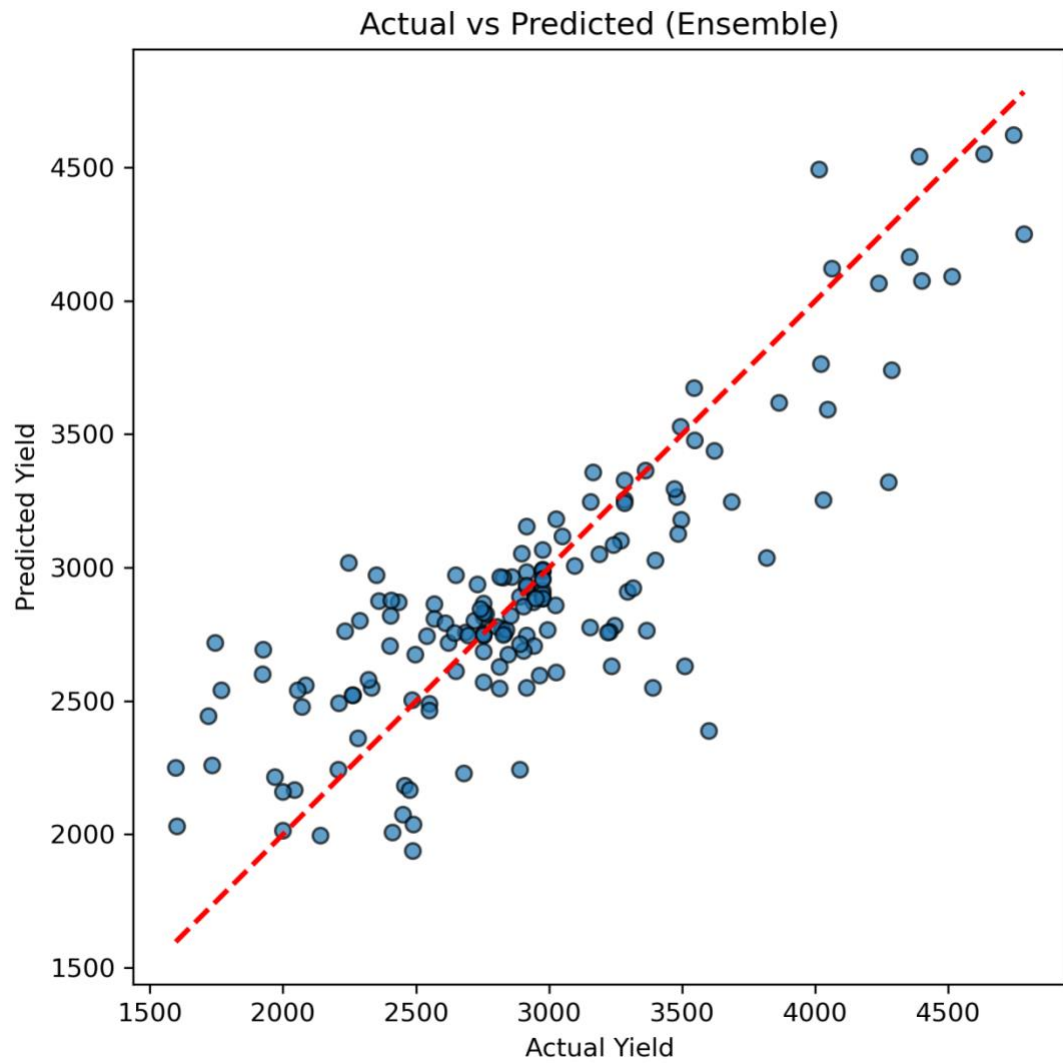


Figure 5.6 Ensemble Model Actual v/s Predicted Plot

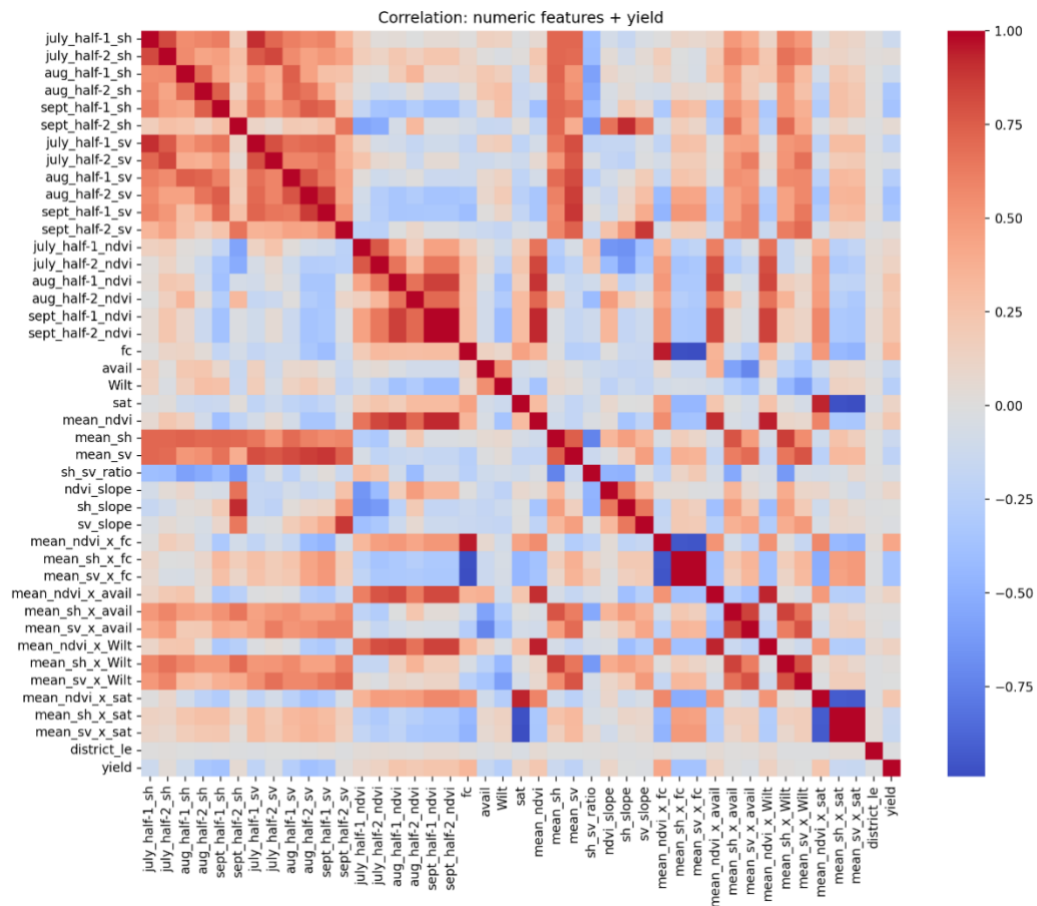


Figure 5.7 Feature Correlation Heatmap

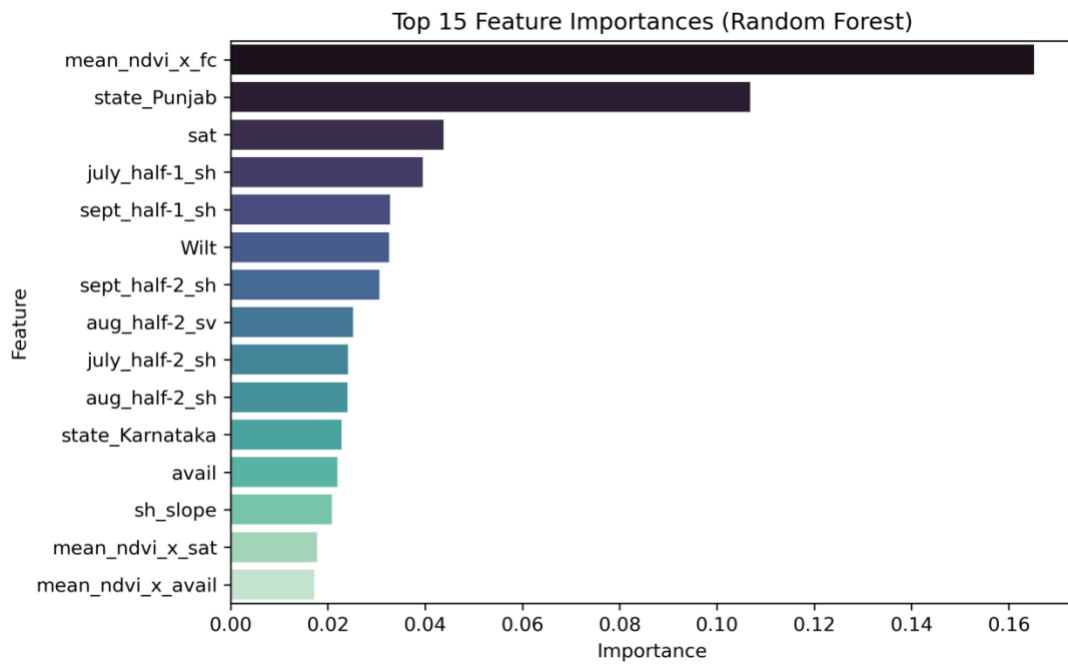


Figure 5.8 RF Feature Importance

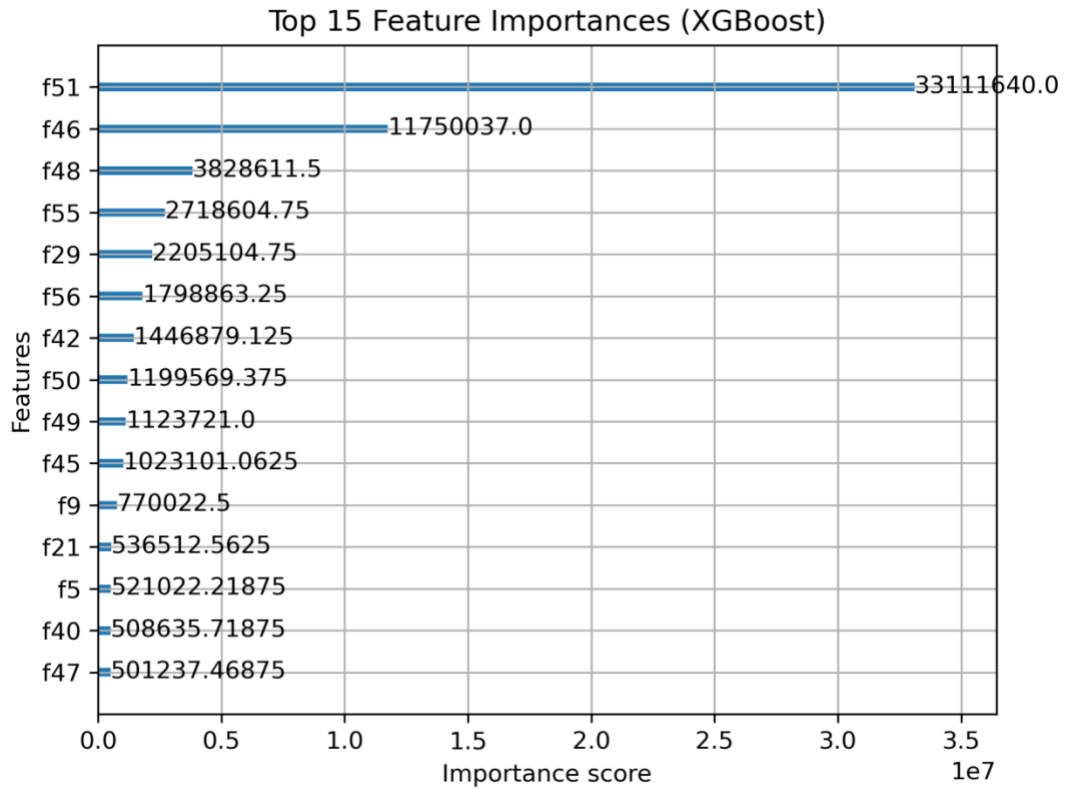


Figure 5.9 XGB Feature Importance

5.6 Discussion

1. **RF vs XGB:** XGB performed slightly better in R^2 initially; RF required tuning to reach comparable accuracy. In the final run, optimized RF model outperformed XGBoost. Ensemble approach leveraged complementary strengths of both models.
2. **Temporal insights:** NDVI during August–September and SH/SV ratios in July–August were consistently the strongest predictors, corresponding to peak vegetative and reproductive phases of rice growth.
3. **State-wise variation:** High rice-density states (Punjab, Haryana) had more predictable yields due to less fragmented fields, whereas fragmented rice regions (Jharkhand, Chhattisgarh) relied more heavily on hybrid gap-filled σ^0 data.
4. **Importance of Ensemble:** Combining RF and XGB reduced prediction errors and stabilized R^2 across folds.
5. **Integration of Multi-Sensor Data:** Incorporating both optical and microwave indices, along with soil parameters, was crucial for achieving accurate district-level predictions.

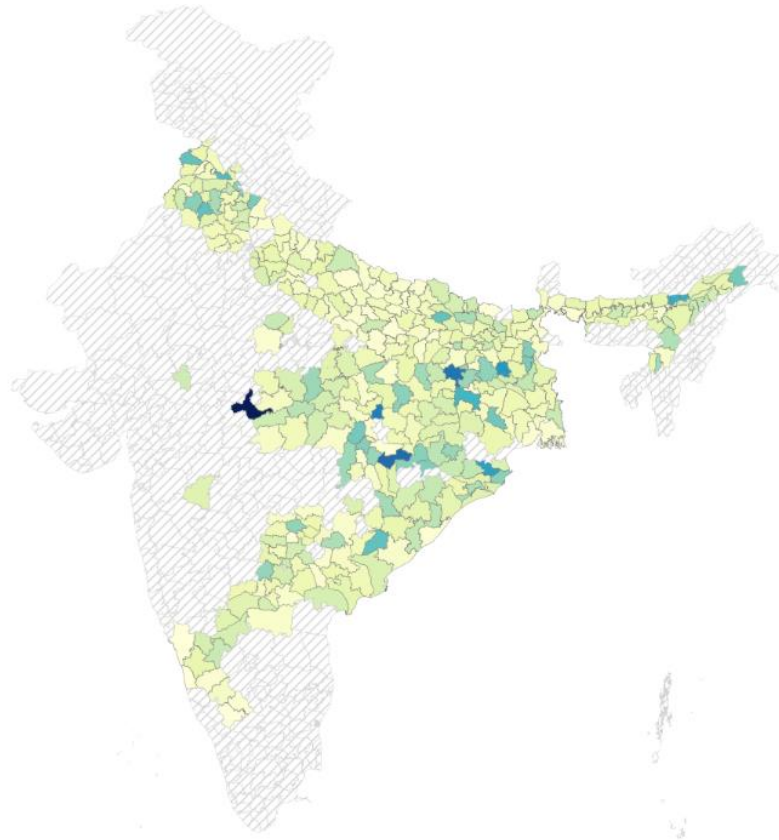


Figure 5.10 District-wise MAE

5.7 Summary

- The **initial models provided a baseline**, showing XGB slightly outperforming RF.
- **Hyperparameter tuning** optimized both models, improving predictive consistency. After which the results favoured RF over XGB.
- **Ensemble predictions** offered the best overall performance ($R^2 = 0.6941$, MAE = 266.006 t/ha).
- Feature analysis confirms the **synergistic value** of combining NDVI, σ^0 , SH/SV ratios, and soil-water parameters in yield prediction.
- The final pipeline provides a **reproducible, scalable, and interpretable framework** for regional rice yield forecasting.

CHAPTER-6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study presented a comprehensive workflow for predicting **district-level rice yield** in India using a combination of **Sentinel-2 optical indices**, **EOS-06 SCAT-3 σ^0 backscatter data**, and **soil parameters**. Key achievements include:

- **Integrated Multi-Source Dataset:** A master dataset was created combining NDVI, σ^0 (SH and SV), SH/SV ratios, and soil-water parameters (AWC, FC, WP, SWC) for 13 states across the Kharif season. Each half-month interval was represented separately, allowing fine temporal resolution for crop growth monitoring.
- **Robust SCAT-3 Processing:** Raw σ^0 values were converted, filtered using rice crop masks, and aggregated to district-level time series. A hybrid gap-filling approach ensured complete and reliable datasets.
- **Feature Engineering:** The SH/SV ratio and temporal aggregation captured key vegetation dynamics. NDVI and EVI integration with soil parameters improved the dataset's predictive capability.
- **Machine Learning Modeling:** Random Forest and XGBoost models were trained on the master dataset, demonstrating the potential of multi-source remote sensing data for accurate rice yield prediction. Preliminary evaluation suggests that both models provide reasonable predictive accuracy, with feature importance highlighting the critical periods and variables influencing yield.

Overall, the methodology illustrates a **scalable, reproducible, and modular framework** that can be applied to other crops, regions, or years with minimal adjustments.

6.2 Future Work

Several opportunities exist to extend and enhance this study:

1. **Additional Temporal Coverage:** Including more years or finer temporal

resolution could improve model robustness and capture inter-annual variability.

2. **Incorporation of Additional Environmental Data:** Rainfall, temperature, and soil moisture from other sensors could further improve prediction accuracy.
3. **Expansion to Other Crops:** The workflow is adaptable to other crops with appropriate crop masks, allowing multi-crop yield prediction.
4. **Advanced Modeling Techniques:** Exploration of deep learning architectures (e.g., LSTM or Temporal CNN) may capture temporal dependencies more effectively.
5. **Spatial Resolution Enhancement:** Integrating higher-resolution Sentinel-2 imagery or finer SCAT-3 grid products could enhance district-level yield estimates.

REFERENCES

1. European Space Agency (ESA). (2024). Sentinel-2 User Handbook: Copernicus Open Access Hub Data Products. Retrieved from <https://scihub.copernicus.eu>
2. Meteorological and Oceanographic Satellite Data Archival Centre (MOSDAC). (2024). EOS-06 (SCAT-3) Scatterometer Level-2 Data Products. Indian Space Research Organisation (ISRO). Retrieved from <https://mosdac.gov.in>
3. Google Earth Engine Developers. (2023). Google Earth Engine Platform for Geospatial Analysis. Retrieved from <https://earthengine.google.com>
4. GADM Database of Global Administrative Areas. (2024). Administrative Boundaries – Level 2 (Districts) for India. Retrieved from <https://gadm.org>
5. Food and Agriculture Organization (FAO). (2023). Soil Water Retention and Field Capacity Parameters for Agroecological Zones. Rome: FAO.
6. Li, Y., Wang, Q., & Zhang, X. (2020). Machine learning approaches for rice yield estimation using multi-source remote sensing data. *Agricultural Systems*, 181, 102808. <https://doi.org/10.1016/j.agsy.2020.102808>
7. Zhang, X., Liu, L., & Xiao, J. (2021). Multi-sensor data fusion for crop yield prediction: Integrating optical, microwave, and climatic variables. *Remote Sensing*, 13(14), 2765. <https://doi.org/10.3390/rs13142765>
8. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
9. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>
10. Indian Space Research Organisation (ISRO). (2023). EOS-06 (Oceansat-3) Mission Overview. Space Applications Centre, Ahmedabad. Retrieved from <https://www.isro.gov.in>
11. Dutta, S., & Bhattacharya, B. (2021). Operational Agricultural Monitoring using Microwave Remote Sensing: Advances and Case Studies. Space Applications Centre (ISRO).
12. Hijmans, R. J., & van Etten, J. (2012). Raster: Geographic data analysis and modeling. R package version 3.4-13.

13. Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2), 127–150.
14. Huete, A. R. (1999). A soil-adjusted vegetation index (SAVI). *Remote Sensing of Environment*, 25(3), 295–309.